CAP 3330

Module 3: ANOVA

Two-way ANOVA

Two-way ANOVA

- We are interested in studying the effect of two factors (i.e., two categorical variables) on one quantitative DV.

For example: The influence of battery brand (factor 1) and the battery material (factor 2) on the battery lifetime (DV).

- Factor A has **a** levels and Factor B has **b** levels.
- We have **a*b** *treatments or experimental conditions*. They are the combinations of the levels between factor A and factor B.

Example: If Factor A is battery brand and Factor B is the battery material, then, the treatments are all possible combinations between brands and types of material. For example, one treatment is Brand A and material 1, another treatment is Brand A and material 2, another treatment is Brand B and material 1, …

Two-way ANOVA

- In two-way ANOVA, we are interested in studying **three effects**: the **individual (or main) effect of each factor** and the **interaction effect**.
- The **individual (or main) effect** of a factor is significant IF the average value of the DV is statistically different for different levels of the factor.

- For example: The main effect of the factor battery brand is significant IF the average lifetime is significantly different for at least two battery brands.

Two way ANOVA

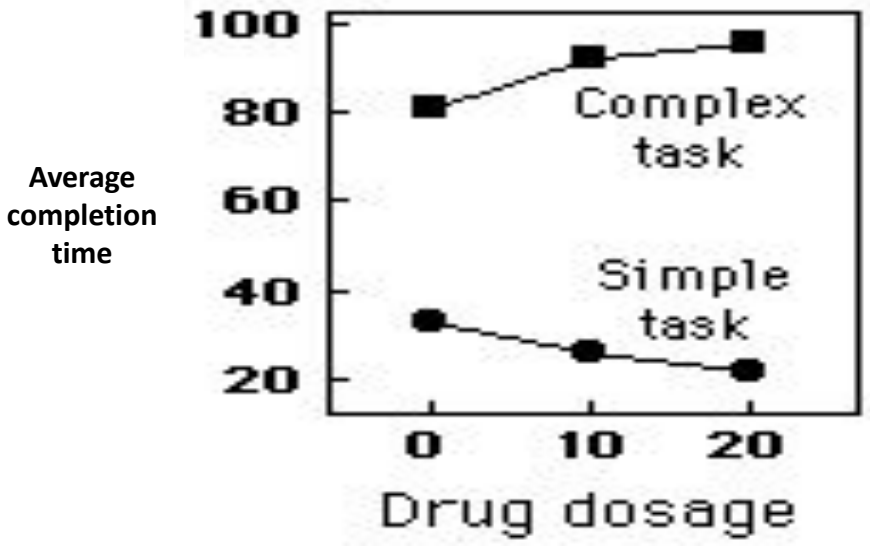
Interaction effect: There is an **interaction effect** between factors A and B **IF** the effect of factor A on the DV **depends** on the level of factor B.

In other words, there is interaction WHEN the effect of factor A on the DV **changes depending** on the level of factor B.

Example: This is a possible manifestation of interaction: changing from **brand A to brand B** produces an ***increase*** in lifetime when **material 1** is used, but changing from **brand A to brand B** produces a ***decrease*** in lifetime when **material 2** is used.

Understanding interaction from an **Interaction Plot**

- **DV (Y axis)**: Participant's task completion time
- **Factor A (X axis)**: Drug dosage (three levels= no drug, 10 mg, and 20 mg)
- **Factor B (Two distinct lines)**: Task complexity (two levels= simple and complex)



Hypotheses in a two-way ANOVA

Main effect of Factor A (A has **a** levels (1,2,...a))

H_0: $\mu_1 = \mu_2 = ... = \mu_a$

Ha: At least two levels of Factor A have statistically different population means for the DV

Main effect of Factor B (B has **b** levels (1,2,...b))

H_0: $\mu_1 = \mu_2 = ... = \mu b$

Ha: At least two levels of Factor B have statistically different population means for the DV

Effect of interaction between A and B

Ho: No interaction effect (= The effect of **factor A** on the DV does NOT depend on the level of **factor B**).

Ha: There is an interaction effect (= The effect of **factor A** on the DV depends on the level of **factor B**).

ANOVA table in a two-way ANOVA (SKIM)

Source of Variation	*Sum of Squares*	*Degrees of Freedom*	*Mean Square*	*F*
Factor A	SS_A	$a-1$	MS_A	F_A
Factor B	SS_B	$b-1$	MS_B	F_B
A*B	SS_{A*B}	$(a-1)(b-1)$	MS_{A*B}	F_{A*B}
Error	SS_E	$ab(n-1)$	MS_E	
Total	SS_{Total}	$abn-1$		

a: # levels of Factor A

b: # levels of Factor B

n: # replicates = # of observations per treatment (assuming it is constant across treatments) (CAREFUL, n isn't sample size)

- In a one-way ANOVA, we had ONLY two sources of variation: the within-group variability (error or residual variation) and the between-group variability (the variability due to the factor).
- We have 4 sources of variation in a two-way ANOVA. Between-group variability due to factor A, between-group variability due to factor B, between-group variability not accounted for the main effects of A and B (= between group variability due to interaction), and within-group variability (variability due to error)

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